

STATICS AND DYNAMICS

Complete Master Lecture Notes & Solved Question Manual

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Mathematically Rigorous, Fully Typeset into LaTeX with TikZ Diagrams by Aryan

Complete 7-Part Syllabus Scope Included in Full Mathematical Rigor:

- **Part I: Forces in Two-Dimensions & Friction:** Coplanar forces, Varignon's Theorem, reduction (X, Y, G) , line of action $xY - yX = G$, Coulomb's friction laws, angle of friction $\lambda = \tan^{-1} \mu$, cone of friction, inclined planes, solved problems.
- **Part II: Principle of Virtual Work & Stability:** Virtual displacement, workless constraints, principle $\sum \delta W = 0$, jointed frameworks, potential energy criterion $V(\theta)$, stability condition $\frac{1}{h} > \frac{1}{r} + \frac{1}{R}$.
- **Part III: Common Catenary & Flexible Strings:** Intrinsic equation $s = c \tan \psi$, Cartesian equation $y = c \cosh(x/c)$, master relations $y^2 = c^2 + s^2$, $T = wy$, $T_0 = wc$, span, sag, parabolic catenaries.
- **Part IV: Kinematics, Rectilinear Motion & SHM:** Cartesian/Polar/Intrinsic velocity and acceleration, SHM ($\ddot{x} = -\mu x$), period $T = \frac{2\pi}{\sqrt{\mu}}$, inverse square law, motion in resisting medium, terminal velocity.
- **Part V: Central Orbits & Kepler's Laws:** Radial/transverse velocity and acceleration, polar differential equation $\frac{d^2 u}{d\theta^2} + u = \frac{P}{h^2 a^2}$, pedal equation $\frac{h^2}{p^3} \frac{dp}{dr} = P$, apsides, Kepler's 1st, 2nd, and 3rd laws, inverse square law orbits.
- **Part VI: Constrained Motion on Curves:** Tangential/normal acceleration, smooth vertical circle motion ($u \geq \sqrt{5ga}$), cycloidal pendulum and isochronous motion.
- **Part VII: Rotating Reference Frames & Coriolis Force:** Transformation equations, centrifugal force, Coriolis force $\mathbf{F}_{\text{Coriolis}} = -2m(\boldsymbol{\omega} \times \mathbf{v}_R)$.
- **Master Formula Cheat Sheet:** Comprehensive formula box on the final page.

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1 Part I: Forces in Two Dimensions & Friction

1.1 Analytical Conditions of Equilibrium of Coplanar Forces

Definition: Coplanar Concurrent Forces

When a system of n forces $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ acts on a particle at point O making angles $\theta_1, \theta_2, \dots, \theta_n$ relative to an X -axis:

$$R_x = \sum_{i=1}^n P_i \cos \theta_i \quad (1)$$

$$R_y = \sum_{i=1}^n P_i \sin \theta_i \quad (2)$$

The magnitude of the resultant force is:

$$R = \sqrt{R_x^2 + R_y^2} \quad (3)$$

The particle is in ****static equilibrium**** ($R = 0$) if and only if:

$$R_x = 0 \quad \text{and} \quad R_y = 0 \quad (4)$$

Theorem / Law: Geometrical Condition of Equilibrium (Polygon Law)

If any number of coplanar forces acting at a point can be represented in magnitude and direction by the sides of a closed polygon taken in order, then the forces will be in static equilibrium.

Theorem / Law: Parallelogram Law of Forces

If two coplanar forces P and Q acting simultaneously on a particle at angle α are represented in magnitude and direction by adjacent sides of a parallelogram, their resultant R is represented by the diagonal passing through that point:

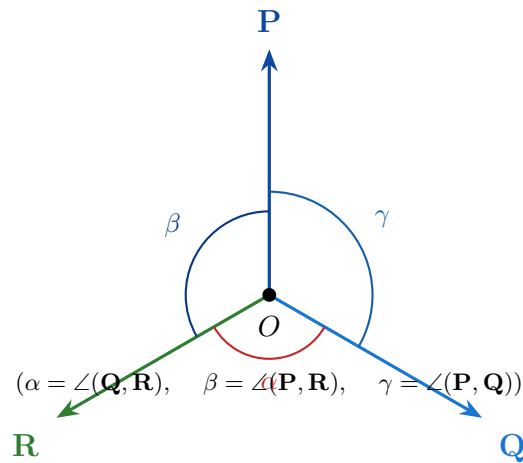
$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}, \quad \tan \theta = \frac{Q \sin \alpha}{P + Q \cos \alpha} \quad (5)$$

Theorem / Law: Lami's Theorem

If three coplanar concurrent forces P, Q, R acting at a point are in static equilibrium, each force is proportional to the sine of the angle between the other two forces:

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma} \quad (6)$$

where $\alpha = \angle(\mathbf{Q}, \mathbf{R})$, $\beta = \angle(\mathbf{P}, \mathbf{R})$, $\gamma = \angle(\mathbf{P}, \mathbf{Q})$.



1.2 Moments and Couples

Definition: Moment of Force About a Point

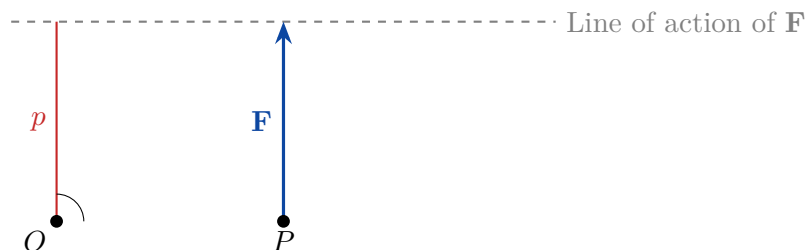
The moment of a force $\mathbf{F}$ acting at a point with position vector $\mathbf{r}$ relative to moment center O is defined as:

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} \quad (7)$$

Scalar magnitude of moment:

$$M = F \cdot p \quad (8)$$

where p is the perpendicular distance from O to the line of action of force $\mathbf{F}$.



Definition: Moment of Force About a Line

Let $\mathbf{F}$ be a force acting at a point and let OZ be any line in space. Let θ be the angle between $\mathbf{F}$ and line OZ . Resolving $\mathbf{F}$ into: 1. $F \cos \theta$ parallel to line OZ 2. $F \sin \theta$ perpendicular to line OZ

The moment of force $\mathbf{F}$ about line OZ is defined as the scalar moment of component $F \sin \theta$ about the point of intersection of line OZ with a plane perpendicular to OZ :

$$M_{OZ} = (\mathbf{r} \times \mathbf{F}) \cdot \hat{\mathbf{k}} \quad (9)$$

Theorem / Law: Varignon's Theorem of Moments

The algebraic sum of the moments of any two coplanar forces about any point in their plane is equal to the moment of their resultant about that same point:

$$\mathbf{M}_O(\mathbf{R}) = \mathbf{M}_O(\mathbf{P}) + \mathbf{M}_O(\mathbf{Q}) \quad (10)$$

Proof / Derivation: Vector Proof of Varignon's Theorem

Let two forces $\mathbf{P}$ and $\mathbf{Q}$ act at point A with position vector $\mathbf{r}$ relative to moment center O . The resultant force $\mathbf{R}$ is $\mathbf{R} = \mathbf{P} + \mathbf{Q}$. Taking the vector cross product with $\mathbf{r}$ from the left:

$$\mathbf{r} \times \mathbf{R} = \mathbf{r} \times (\mathbf{P} + \mathbf{Q}) = (\mathbf{r} \times \mathbf{P}) + (\mathbf{r} \times \mathbf{Q})$$

This proves that the vector moment of the resultant equals the sum of the vector moments of the individual forces.

Theorem / Law: Theory of Couples

A **couple** consists of two equal, opposite, and non-collinear parallel forces ($P, -P$). 1. The vector sum of forces in a couple is zero ($\sum \mathbf{F} = \mathbf{0}$). A couple produces pure rotation. 2. The moment of a couple $G = P \cdot d$ (where d is arm of couple) is **constant** and independent of the base point. 3. The resultant of any number of coplanar couples of moments $G_1, G_2, \dots, G_n$ is a single couple of moment $G = \sum_{i=1}^n G_i$.

1.3 Reduction of General Coplanar System of Forces**Theorem / Law: Reduction Theorem for Coplanar Forces**

A system of coplanar forces $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ acting at points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ on a rigid body can be reduced to a single resultant force (X, Y) acting at an arbitrary origin O , together with a single couple of moment G about O :

$$X = \sum_{i=1}^n X_i, \quad Y = \sum_{i=1}^n Y_i \quad (11)$$

$$G = \sum_{i=1}^n (x_i Y_i - y_i X_i) \quad (12)$$

where (X_i, Y_i) are the components of force $\mathbf{P}_i$. Resultant Force Magnitude: $R = \sqrt{X^2 + Y^2}$ and direction $\tan \theta = Y/X$.

Proof / Derivation: Proof of Reduction Theorem

Let force $\mathbf{P}_i$ act at point $A_i(x_i, y_i)$ with components (X_i, Y_i) . Transferring component X_i to origin $O(0, 0)$ introduces: 1. Equal force X_i at O 2. Couple of moment $-y_i X_i$

Transferring component Y_i to origin $O(0, 0)$ introduces: 1. Equal force Y_i at O 2. Couple of moment $+x_i Y_i$

Summing over all n forces: - Total force along X -axis at O : $X = \sum X_i$ - Total force along Y -axis at O : $Y = \sum Y_i$ - Total couple moment about O : $G = \sum (x_i Y_i - y_i X_i)$

Theorem / Law: Line of Action of Resultant Force

If $R \neq 0$, the equation of the line of action of the resultant force R relative to origin O is:

$$xY - yX = G \quad (13)$$

Proof / Derivation: Proof of Line of Action Equation

Let the base point be shifted from origin $O(0,0)$ to a point $O'(\xi, \eta)$. The force components remain (X, Y) , but the new couple moment about O' becomes:

$$G' = G - \xi Y + \eta X$$

For $O'(\xi, \eta)$ to lie on the line of action of resultant force R , the couple moment G' must vanish ($G' = 0$):

$$G - \xi Y + \eta X = 0 \implies \xi Y - \eta X = G$$

Replacing coordinates (ξ, η) with standard rectangular coordinates (x, y) gives:

$$xY - yX = G$$

1.4 Laws of Friction and Equilibrium on Rough Surfaces**Definition: Friction and Limiting Friction**

1. **Friction:** The tangential force exerted by a rough surface on a body resting on it, opposing relative motion. 2. **Laws of Limiting Friction:** - Limiting friction $F_{\max} = \mu N$, where μ is the coefficient of static friction and N is normal reaction. - Angle of Friction $\lambda = \tan^{-1} \mu$. - Cone of Friction: Right circular cone of semi-vertical angle $\lambda = \tan^{-1} \mu$ with axis along normal reaction N .

Solved Question / Problem: Equilibrium of Body on Rough Inclined Plane

A heavy body of weight W rests on a rough inclined plane of inclination α to horizontal. Find the minimum horizontal force P required to prevent the body from sliding down the plane.

Proof / Derivation: Solution

Let λ be the angle of friction ($\mu = \tan \lambda$). When the body is on the verge of sliding down, friction $F = \mu N$ acts up the plane. Resolving forces along and perpendicular to the inclined plane:

$$P \cos \alpha + F = W \sin \alpha \implies F = W \sin \alpha - P \cos \alpha \quad (14)$$

$$N = W \cos \alpha + P \sin \alpha \quad (15)$$

Setting limiting friction $F = \mu N = N \tan \lambda$:

$$W \sin \alpha - P \cos \alpha = (W \cos \alpha + P \sin \alpha) \frac{\sin \lambda}{\cos \lambda}$$

Cross-multiplying by $\cos \lambda$:

$$W(\sin \alpha \cos \lambda - \cos \alpha \sin \lambda) = P(\cos \alpha \cos \lambda + \sin \alpha \sin \lambda)$$

$$W \sin(\alpha - \lambda) = P \cos(\alpha - \lambda) \implies P = W \tan(\alpha - \lambda)$$

1.5 Solved Standard Exam Problems from Part I**Solved Question / Problem: Forces Along Sides of Triangle AB, BC, CA**

Forces equal to $3P, 7P, 5P$ act along the sides AB, BC, CA of an equilateral triangle ABC of side a , taken in order. Find the magnitude, direction, and line of action of the resultant force.

Proof / Derivation: Solution

Let side AB lie along the X -axis with A at origin $(0,0)$. Side angles with X -axis: 0° for AB , 120° for BC , 240° for CA .

1. **Horizontal Component X** :

$$X = 3P \cos 0^\circ + 7P \cos 120^\circ + 5P \cos 240^\circ = 3P + 7P \left(-\frac{1}{2}\right) + 5P \left(-\frac{1}{2}\right) = -3P$$

2. **Vertical Component Y** :

$$Y = 3P \sin 0^\circ + 7P \sin 120^\circ + 5P \sin 240^\circ = 0 + 7P \left(\frac{\sqrt{3}}{2}\right) + 5P \left(-\frac{\sqrt{3}}{2}\right) = \sqrt{3}P$$

3. **Resultant Magnitude and Direction**:

$$R = \sqrt{X^2 + Y^2} = \sqrt{(-3P)^2 + (\sqrt{3}P)^2} = \sqrt{12P^2} = 2\sqrt{3}P$$

$$\tan \theta = \frac{Y}{X} = \frac{\sqrt{3}P}{-3P} = -\frac{1}{\sqrt{3}} \Rightarrow \theta = 150^\circ$$

4. **Resultant Moment G_A about A** : Taking moments about vertex $A(0,0)$: Forces $3P$ and $5P$ pass through $A \Rightarrow$ moments = 0. Perpendicular distance from A to line BC is $p = a \sin 60^\circ = \frac{\sqrt{3}}{2}a$.

$$G_A = 7P \cdot \left(\frac{\sqrt{3}}{2}a\right) = \frac{7\sqrt{3}}{2}Pa$$

Line of action equation $xY - yX = G_A$:

$$x(\sqrt{3}P) - y(-3P) = \frac{7\sqrt{3}}{2}Pa \Rightarrow \sqrt{3}x + 3y = \frac{7\sqrt{3}}{2}a \Rightarrow 2x + 2\sqrt{3}y = 7a$$

Solved Question / Problem: Forces Along Sides of Regular Hexagon

$ABCDEF$ is a regular hexagon of side a . Forces $P, 2P, 3P, 4P, 5P, 6P$ act along sides AB, BC, CD, DE, EF, FA respectively. Find the magnitude, direction, and line of action of the resultant.

Proof / Derivation: Solution

Let side AB lie along X -axis with A at origin $(0,0)$. Angles of sides with X -axis: $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$.

1. **Resolution**:

$$X = P \cos 0^\circ + 2P \cos 60^\circ + 3P \cos 120^\circ + 4P \cos 180^\circ + 5P \cos 240^\circ + 6P \cos 300^\circ = -3P$$

$$Y = P \sin 0^\circ + 2P \sin 60^\circ + 3P \sin 120^\circ + 4P \sin 180^\circ + 5P \sin 240^\circ + 6P \sin 300^\circ = -3\sqrt{3}P$$

$$R = \sqrt{(-3P)^2 + (-3\sqrt{3}P)^2} = 6P, \quad \tan \theta = \sqrt{3} \Rightarrow \theta = 240^\circ$$

2. **Moment about Center O** : Perpendicular distance to any side is $p = \frac{\sqrt{3}}{2}a$.

$$G_O = (P + 2P + 3P + 4P + 5P + 6P) \frac{\sqrt{3}}{2}a = 21P \cdot \frac{\sqrt{3}}{2}a = \frac{21\sqrt{3}}{2}Pa$$

2 Part II: Principle of Virtual Work & Stability

2.1 Principle of Virtual Work

Theorem / Law: Principle of Virtual Work

A system of coplanar forces acting on a rigid body or interconnected system with ideal workless constraints is in **static equilibrium** if and only if the total virtual work done by all external forces in any arbitrary virtual displacement consistent with constraints vanishes:

$$\sum \delta W = 0 \quad (16)$$

Proof / Derivation: Virtual Work for Common Constraint Forces

1. **Extensible / Inextensible String**: $\delta W = -T\delta s$. For inextensible string ($\delta s = 0$), $\delta W = 0$.
2. **Light Rigid Rod**: $\delta W = -R\delta l$. For rigid rod ($\delta l = 0$), $\delta W = 0$.
3. **Smooth Surface Reaction**: Normal reaction $N \perp \delta \mathbf{r} \implies \delta W = N \cdot 0 = 0$.
4. **Fixed Smooth Hinge**: Point of application does not move $\implies \delta W = 0$.
5. **Gravity Force**: $\delta W = -W\delta y_{cg}$.

2.2 Solved Examples on Jointed Frameworks

Solved Question / Problem: Four Jointed Uniform Rods Forming Rhombus

Four equal uniform rods, each of weight w and length a , are freely jointed to form a rhombus $ABCD$. The framework is suspended vertically from vertex A . A weight W is attached at the lowest vertex C , and the framework is kept in shape by a light inextensible string joining B and D . If $\angle DAB = 2\theta$, find the tension T in string BD .

Proof / Derivation: Solution using Virtual Work

Let A be the fixed origin $(0,0)$ with Y -axis vertically downwards. 1. **Coordinates**: - Depth of vertex C : $y_C = 2a \cos \theta$. - Depth of center of gravity of 4 rods: $y_{cg} = a \cos \theta$. Total weight of 4 rods $= 4w$. - Length of diagonal BD : $x = 2a \sin \theta$.

2. **Virtual Work Equation**: Active forces are weight W at C , weight $4w$ at center of gravity, and tension T along BD :

$$\delta W = W\delta y_C + 4w\delta y_{cg} - T\delta x = 0 \quad (17)$$

3. **Differentiating Coordinates**: - $\delta y_C = -2a \sin \theta \delta \theta$ - $\delta y_{cg} = -a \sin \theta \delta \theta$ - $\delta x = 2a \cos \theta \delta \theta$

4. **Substitution**:

$$W(-2a \sin \theta \delta \theta) + 4w(-a \sin \theta \delta \theta) - T(2a \cos \theta \delta \theta) = 0$$

Dividing by $-2a\delta\theta$:

$$W \sin \theta + 2w \sin \theta + T \cos \theta = 0 \implies T = (2w + W) \tan \theta$$

Solved Question / Problem: Six Jointed Equal Rods Forming Regular Hexagon

Six equal rods AB, BC, CD, DE, EF, FA , each of weight w , are jointed together to form a regular hexagon $ABCDEF$. The framework is suspended vertically from vertex A , and a weight W is attached at lowest vertex D . A light horizontal string joins the midpoints of AB and AF . Find the tension T in the string using Virtual Work.

Proof / Derivation: Solution

Let θ be the angle each upper rod makes with the vertical diagonal AD . For a regular hexagon, $\theta = 30^\circ$. 1. ****Coordinates****: - Depth of lowest vertex D : $y_D = 4a \cos \theta$. - Depth of center of gravity of 6 rods: $y_{cg} = 2a \cos \theta$. Total weight = $6w$. - Distance between midpoints of AB and AF : $x = a \sin \theta$.

2. ****Virtual Work Equation****:

$$\delta W = W\delta y_D + 6w\delta y_{cg} - T\delta x = 0$$

$$W(-4a \sin \theta \delta \theta) + 6w(-2a \sin \theta \delta \theta) - T(a \cos \theta \delta \theta) = 0$$

Dividing by $a\delta\theta$:

$$T \cos \theta = (4W + 12w) \sin \theta \implies T = (4W + 12w) \tan \theta$$

For $\theta = 30^\circ$, $\tan 30^\circ = \frac{1}{\sqrt{3}}$:

$$T = \frac{4W + 12w}{\sqrt{3}} = \frac{4}{\sqrt{3}}(W + 3w) \quad (18)$$

2.3 Stability of Equilibrium and Potential Energy Criteria

Theorem / Law: Potential Energy Criterion for Stability

For conservative systems with potential energy $V(\theta)$: - ****Equilibrium****: $\frac{dV}{d\theta} = 0$ - ****Stable Equilibrium****: $\frac{d^2V}{d\theta^2} > 0$ (Local Minimum of V) - ****Unstable Equilibrium****: $\frac{d^2V}{d\theta^2} < 0$ (Local Maximum of V)

Theorem / Law: Stability Criterion for Curved Surfaces

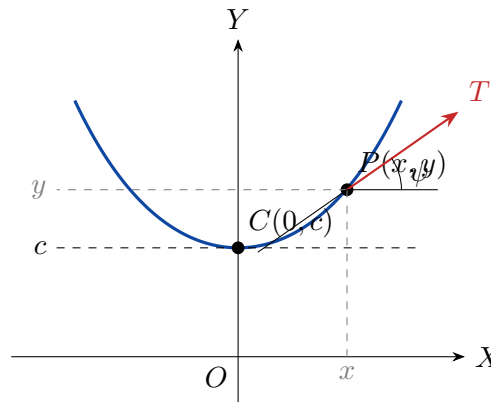
If a body of radius of curvature r rests on top of a fixed convex surface of radius of curvature R with center of gravity at height h above contact point: - ****Stable****: $\frac{1}{h} > \frac{1}{r} + \frac{1}{R}$ - ****Unstable****: $\frac{1}{h} < \frac{1}{r} + \frac{1}{R}$

3 Part III: Common Catenary & Flexible Strings

3.1 Intrinsic and Cartesian Equations of Common Catenary

Definition: Common Catenary

A **common catenary** is the curve formed by a uniform, flexible, inextensible string hanging freely under gravity between two fixed points.


Proof / Derivation: Derivation of Intrinsic Equation $s = c \tan \psi$

Consider an arc CP of length s measured from lowest point $C(0, c)$ to point $P(x, y)$. Let w be the weight per unit length of string. Three forces keep arc CP in equilibrium: 1. Horizontal tension T_0 at lowest point C . 2. Tension T at P acting along tangent at angle ψ to horizontal. 3. Weight of arc CP , $W = ws$, acting vertically downwards.

Resolving forces:

$$T \cos \psi = T_0 \quad (19)$$

$$T \sin \psi = ws \quad (20)$$

Dividing (20) by (19):

$$\tan \psi = \frac{ws}{T_0} \implies s = \left(\frac{T_0}{w} \right) \tan \psi = c \tan \psi$$

where parameter $c = \frac{T_0}{w}$.

Proof / Derivation: Derivation of Cartesian Equation $y = c \cosh(x/c)$

From calculus, $\frac{dy}{dx} = \tan \psi$ and $\frac{ds}{dx} = \sec \psi$. Differentiating $s = c \tan \psi$ with respect to x :

$$\frac{ds}{dx} = c \sec^2 \psi \frac{d\psi}{dx} \implies \sec \psi = c \sec^2 \psi \frac{d\psi}{dx} \implies dx = c \sec \psi d\psi$$

Integrating $dx = c \sec \psi d\psi$:

$$x = c \ln(\sec \psi + \tan \psi) \implies \sec \psi + \tan \psi = e^{x/c}$$

Since $\sec \psi - \tan \psi = e^{-x/c}$, adding gives $2 \sec \psi = e^{x/c} + e^{-x/c} \implies \sec \psi = \cosh(x/c)$.

Integrating $\frac{dy}{dx} = \tan \psi = \sinh(x/c)$ with $y(0) = c$:

$$y = c \cosh\left(\frac{x}{c}\right) \quad (21)$$

Important Key Note / Exam Point: Master Relations for Common Catenary

$$y^2 = c^2 + s^2 \quad (22)$$

$$T = wy \quad (23)$$

$$T_0 = wc \quad (24)$$

$$x = c \ln(\sec \psi + \tan \psi) = c \sinh^{-1} \left(\frac{s}{c} \right) \quad (25)$$

3.2 Solved Examples on Catenary**Solved Question / Problem: Tension at Any Point of Catenary**

Prove that the tension T at any point $P(x, y)$ of a common catenary is equal to the weight of a length of the string equal to the height of P above the directrix ($T = wy$).

Proof / Derivation: Proof

Squaring and adding equations (19) and (20):

$$T^2(\cos^2 \psi + \sin^2 \psi) = T_0^2 + w^2 s^2 = w^2 c^2 + w^2 s^2 = w^2(c^2 + s^2)$$

Using the master relation $c^2 + s^2 = y^2$:

$$T^2 = w^2 y^2 \implies T = wy$$

This completes the proof.

Solved Question / Problem: Length of String Given Span and Sag

A uniform string of length l hangs between two points in the same horizontal line. If the span is $2b$ and sag is d , show that:

$$y_A = c + d, \quad (c + d)^2 = c^2 + \left(\frac{l}{2}\right)^2 \implies c = \frac{l^2 - 4d^2}{8d} \quad (26)$$

Proof / Derivation: Solution

At the support point $A(b, y_A)$, half-length of string is $s = l/2$. Height of A above directrix: $y_A = c + d$. Using $y^2 = c^2 + s^2$:

$$(c + d)^2 = c^2 + \left(\frac{l}{2}\right)^2 \implies c^2 + 2cd + d^2 = c^2 + \frac{l^2}{4}$$

$$2cd + d^2 = \frac{l^2}{4} \implies 2cd = \frac{l^2 - 4d^2}{4} \implies c = \frac{l^2 - 4d^2}{8d}$$

4 Part IV: Kinematics, Rectilinear Motion & Simple Harmonic Motion

4.1 Kinematics in 2D Systems

Theorem / Law: Expressions for Velocity and Acceleration

1. ****Cartesian Coordinates (x, y) ****: - Velocity: $\mathbf{v} = \dot{x}\hat{\mathbf{i}} + \dot{y}\hat{\mathbf{j}}$, magnitude $v = \sqrt{\dot{x}^2 + \dot{y}^2}$ - Acceleration: $\mathbf{a} = \ddot{x}\hat{\mathbf{i}} + \ddot{y}\hat{\mathbf{j}}$, magnitude $a = \sqrt{\ddot{x}^2 + \ddot{y}^2}$
2. ****Polar Coordinates (r, θ) ****: - Radial velocity: $v_r = \dot{r}$ - Transverse velocity: $v_\theta = r\dot{\theta}$ - Radial acceleration: $a_r = \ddot{r} - r\dot{\theta}^2$ - Transverse acceleration: $a_\theta = \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta}) = r\ddot{\theta} + 2\dot{r}\dot{\theta}$
3. ****Intrinsic Coordinates (s, ψ) ****: - Tangential velocity: $v_t = \dot{s}$ - Normal velocity: $v_n = 0$ - Tangential acceleration: $a_t = \frac{dv}{dt} = v \frac{dv}{ds} = \ddot{s}$ - Normal acceleration: $a_n = \frac{v^2}{\rho}$ (where $\rho = \frac{ds}{d\psi}$)

4.2 Simple Harmonic Motion (SHM)

Definition: Simple Harmonic Motion

A particle moves in a straight line with ****Simple Harmonic Motion**** if its acceleration is always directed towards a fixed point O on the line and is directly proportional to its displacement x from O :

$$\ddot{x} = -\mu x \quad (\mu > 0) \quad (27)$$

Proof / Derivation: Derivation of Velocity and Position in SHM

Writing $\ddot{x} = v \frac{dv}{dx} = -\mu x$:

$$v dv = -\mu x dx \implies \frac{v^2}{2} = -\mu \frac{x^2}{2} + C_1$$

If particle comes to rest at maximum displacement $x = a$ (Amplitude):

$$0 = -\frac{1}{2}\mu a^2 + C_1 \implies C_1 = \frac{1}{2}\mu a^2$$

$$v^2 = \mu(a^2 - x^2) \implies v = \pm \sqrt{\mu} \sqrt{a^2 - x^2} \quad (28)$$

For motion starting from rest at $x = a$:

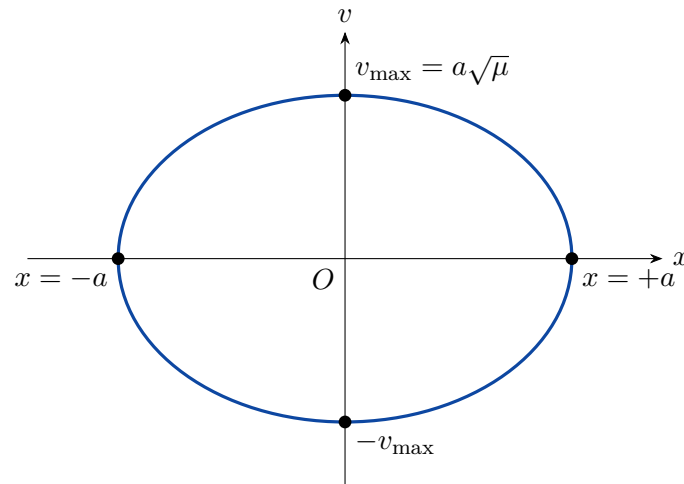
$$\frac{dx}{dt} = -\sqrt{\mu} \sqrt{a^2 - x^2} \implies \frac{dx}{\sqrt{a^2 - x^2}} = -\sqrt{\mu} dt$$

Integrating with $x(0) = a$:

$$\cos^{-1}\left(\frac{x}{a}\right) = \sqrt{\mu} t \implies x(t) = a \cos(\sqrt{\mu} t) \quad (29)$$

****Periodic Time T ****:

$$T = \frac{2\pi}{\sqrt{\mu}} \quad (30)$$



Solved Question / Problem: Velocities at Two Positions in SHM

A point executing SHM has velocities v_1 and v_2 at distances x_1 and x_2 from the center of force. Find the amplitude a and period T of the motion.

Proof / Derivation: Solution

Using $v^2 = \mu(a^2 - x^2)$ at the two given positions:

$$v_1^2 = \mu(a^2 - x_1^2) \quad (31)$$

$$v_2^2 = \mu(a^2 - x_2^2) \quad (32)$$

Subtracting (32) from (31):

$$v_1^2 - v_2^2 = \mu(x_2^2 - x_1^2) \implies \mu = \frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}$$

Periodic time T :

$$T = \frac{2\pi}{\sqrt{\mu}} = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}} \quad (33)$$

To find amplitude a , multiply (31) by x_2^2 and (32) by x_1^2 and subtract:

$$v_1^2 x_2^2 - v_2^2 x_1^2 = \mu a^2 (x_2^2 - x_1^2) = (v_1^2 - v_2^2) a^2 \implies a = \sqrt{\frac{v_1^2 x_2^2 - v_2^2 x_1^2}{v_1^2 - v_2^2}}$$

4.3 Motion in Resisting Medium

Theorem / Law: Motion Under Resistance Proportional to Velocity

If a body of mass m falls vertically under gravity in a medium with resistance mkv : Equation of motion: $\ddot{x} = g - kv$. Setting $\ddot{x} = 0$ gives the **Terminal Velocity V **:

$$V = \frac{g}{k} \quad (34)$$

5 Part V: Central Orbits & Kepler's Laws

5.1 Differential Equations of Central Orbits

Theorem / Law: Differential Equation in $u - \theta$ Form

Let a particle move under a central attractive force $P(r)$ per unit mass towards pole O . Since transverse acceleration $a_\theta = 0 \implies r^2\dot{\theta} = h = \text{constant}$ (Areal velocity $\frac{1}{2}h = \text{constant}$). Substituting $u = 1/r$ into radial equation $\ddot{r} - r\dot{\theta}^2 = -P$ yields the standard differential equation:

$$\frac{d^2u}{d\theta^2} + u = \frac{P}{h^2u^2} \quad (35)$$

Theorem / Law: Pedal Equation ($p - r$) of Central Orbit

In pedal coordinates (p, r) where $p = r \sin \phi = r^2 \frac{d\theta}{ds}$:

$$\frac{h^2}{p^3} \frac{dp}{dr} = P \quad (36)$$

5.2 Kepler's Laws of Planetary Motion

Theorem / Law: Kepler's Laws

1. ****First Law (Law of Orbits)****: Every planet moves in an ellipse with the Sun at one focus.
2. ****Second Law (Law of Areas)****: Radius vector from Sun sweeps equal areas in equal times ($\frac{dA}{dt} = \frac{1}{2}h = \text{constant}$).
3. ****Third Law (Law of Periods)****: Period T satisfies:

$$T^2 = \frac{4\pi^2}{\mu} a^3 \implies T^2 \propto a^3 \quad (37)$$

5.3 Solved Examples on Central Force Laws

Solved Question / Problem: Law of Force for Central Elliptic Orbit

Find the law of force towards the focus for a particle describing an ellipse $\frac{1}{r} = 1 + e \cos \theta$.

Proof / Derivation: Solution

Writing $u = \frac{1}{r} = \frac{1}{l}(1 + e \cos \theta)$. Differentiating with respect to θ :

$$\frac{du}{d\theta} = -\frac{e}{l} \sin \theta \implies \frac{d^2u}{d\theta^2} = -\frac{e}{l} \cos \theta$$

Substituting into differential equation $\frac{d^2u}{d\theta^2} + u = \frac{P}{h^2u^2}$:

$$-\frac{e}{l} \cos \theta + \frac{1}{l}(1 + e \cos \theta) = \frac{P}{h^2u^2} \implies \frac{1}{l} = \frac{P}{h^2u^2}$$

Solving for central force P :

$$P = \frac{h^2}{l} u^2 = \frac{\mu}{r^2} \quad \text{where } \mu = \frac{h^2}{l} \quad (38)$$

Thus the central force obeys the ****Inverse Square Law****.

Solved Question / Problem: Law of Force for Equiangular Spiral

Find the law of force for a particle describing an equiangular spiral $r = ae^{\theta \cot \alpha}$ under a central force towards the pole.

Proof / Derivation: Solution

Pedal equation of equiangular spiral is $p = r \sin \alpha$. Differentiating with respect to r : $\frac{dp}{dr} = \sin \alpha$. Substituting into pedal differential equation $\frac{h^2}{p^3} \frac{dp}{dr} = P$:

$$P = \frac{h^2}{(r \sin \alpha)^3} \sin \alpha = \frac{h^2}{\sin^2 \alpha} \cdot \frac{1}{r^3} = \frac{\mu}{r^3}$$

Thus the force varies inversely as the ****cube**** of the distance.

6 Part VI: Constrained Motion on Curves**6.1 Motion on a Smooth Vertical Circle****Solved Question / Problem: Complete Vertical Loop Condition**

A particle of mass m is projected horizontally with velocity u from the lowest point of a smooth vertical circle of radius a . Find the minimum velocity u required for the particle to complete the full vertical circle.

Proof / Derivation: Solution

Let θ be the angle of radius vector with lowest vertical. By Conservation of Energy:

$$\frac{1}{2}mv^2 + mga(1 - \cos \theta) = \frac{1}{2}mu^2 \implies v^2 = u^2 - 2ga(1 - \cos \theta)$$

Normal reaction N :

$$N - mg \cos \theta = \frac{mv^2}{a} \implies N = \frac{m}{a} [u^2 - 2ga + 3ga \cos \theta]$$

At highest point $\theta = 180^\circ$ ($\cos 180^\circ = -1$):

$$v_{\top}^2 = u^2 - 4ga, \quad N_{\top} = \frac{m}{a}(u^2 - 5ga)$$

To complete the loop without leaving the circle, $N_{\top} \geq 0$:

$$u^2 - 5ga \geq 0 \implies u \geq \sqrt{5ga}$$

6.2 Motion on a Smooth Cycloid

Theorem / Law: Cycloidal Pendulum

For a particle constrained to move under gravity along a smooth cycloid $s = 4a \sin \psi$: Equation of motion: $\ddot{s} = -g \sin \psi = -\frac{g}{4a}s$. This is exact ****Simple Harmonic Motion**** with periodic time:

$$T = 2\pi\sqrt{\frac{4a}{g}} = 4\pi\sqrt{\frac{a}{g}} \quad (39)$$

Since period T is independent of amplitude, motion on a cycloid is ****isochronous****.

7 Part VII: Rotating Reference Frames & Coriolis Force

7.1 Coriolis Acceleration and Coriolis Force

Theorem / Law: Transformation to Rotating Frame

If a frame of reference rotates with constant angular velocity $\boldsymbol{\omega}$ relative to an inertial frame, the relationship between time derivatives of vector $\mathbf{A}$ in fixed frame (F) and rotating frame (R) is:

$$\left(\frac{d\mathbf{A}}{dt}\right)_F = \left(\frac{d\mathbf{A}}{dt}\right)_R + \boldsymbol{\omega} \times \mathbf{A} \quad (40)$$

Applying this transformation twice to position vector $\mathbf{r}$ yields the ****Apparent Acceleration****:

$$\mathbf{a}_F = \mathbf{a}_R + 2(\boldsymbol{\omega} \times \mathbf{v}_R) + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (41)$$

Multiplying by mass m gives Newton's Second Law in rotating frame:

$$m\mathbf{a}_R = \mathbf{F}_{\text{real}} - \underbrace{2m(\boldsymbol{\omega} \times \mathbf{v}_R)}_{\text{Coriolis Force}} - \underbrace{m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})}_{\text{Centrifugal Force}} \quad (42)$$

8 Master Formula Cheat Sheet (Exam Reference)

Important Key Note / Exam Point: Master Formula Reference Cheat Sheet

1. Coplanar Resultant:

$$R = \sqrt{R_x^2 + R_y^2}, \quad \tan \theta = \frac{R_y}{R_x}, \quad xY - yX = G$$

2. Lami's Theorem:

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

3. Limiting Friction & Angle of Friction:

$$F_{\max} = \mu N, \quad \tan \lambda = \mu, \quad P_{\min} = W \tan(\alpha - \lambda)$$

4. Principle of Virtual Work:

$$\sum \delta W = 0 \quad (\text{Strings: } -T\delta s, \text{ Rods: } -R\delta l, \text{ Gravity: } -W\delta y_{cg})$$

5. Equilibrium Stability:

$$\frac{dV}{d\theta} = 0 \implies \text{Equilibrium}, \quad \frac{d^2V}{d\theta^2} > 0 \implies \text{Stable}, \quad \frac{1}{h} > \frac{1}{r} + \frac{1}{R}$$

6. Common Catenary:

$$s = c \tan \psi, \quad y = c \cosh(x/c), \quad y^2 = c^2 + s^2, \quad T = wy, \quad T_0 = wc$$

7. Kinematics & SHM:

$$v_r = \dot{r}, \quad v_\theta = r\dot{\theta}, \quad a_r = \ddot{r} - r\dot{\theta}^2, \quad a_\theta = \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta})$$

$$\ddot{x} = -\mu x \implies v^2 = \mu(a^2 - x^2), \quad T = \frac{2\pi}{\sqrt{\mu}}, \quad x(t) = a \cos(\sqrt{\mu}t)$$

8. Central Orbits & Kepler:

$$\frac{d^2u}{d\theta^2} + u = \frac{P}{h^2u^2}, \quad \frac{h^2}{p^3} \frac{dp}{dr} = P, \quad T^2 = \frac{4\pi^2}{\mu} a^3$$

9. Constrained Vertical Motion:

$$v^2 = u^2 - 2ga(1 - \cos \theta), \quad N = \frac{m}{a}[u^2 - 2ga + 3ga \cos \theta], \quad u_{\text{loop}} \geq \sqrt{5ga}$$

$$\text{Cycloid: } s = 4a \sin \psi \implies \ddot{s} = -\frac{g}{4a}s \implies T = 4\pi \sqrt{\frac{a}{g}}$$

10. Coriolis & Centrifugal Forces:

$$\mathbf{F}_{\text{Coriolis}} = -2m(\boldsymbol{\omega} \times \mathbf{v}_R), \quad \mathbf{F}_{\text{centrifugal}} = -m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$